Title : Product Placement Analysis – Sales Impact

Introduction :

This project analyzes how product placement strategies impact sales performance .The objective is to identify which placement techniques maximize revenue and improve product visibility effectiveness.

Data Preprocessing Steps :

Step 1: Data Cleaning

- Removed duplicate records
- Standardized column names
- Converted date column to date time format
- Corrected inconsistent categorical values

Step 2: Handling Missing Values

- Numeric columns → filled with median
- Categorical columns → filled with mode
- Verified no null values rema

Step 3: Data Type Conversion

- Converted Sales and Units Sold to numeric
- Converted Date column to DateTime

Step 4: Feature Engineering

Created new analytical fields:

- Total Sales Contribution %
- Average Sales per Unit
- Placement-wise Sales Share
- Monthly Sales Trend

Step 5: Outlier Treatment

- Applied IQR method
- Capped extreme values to avoid visualization distortion

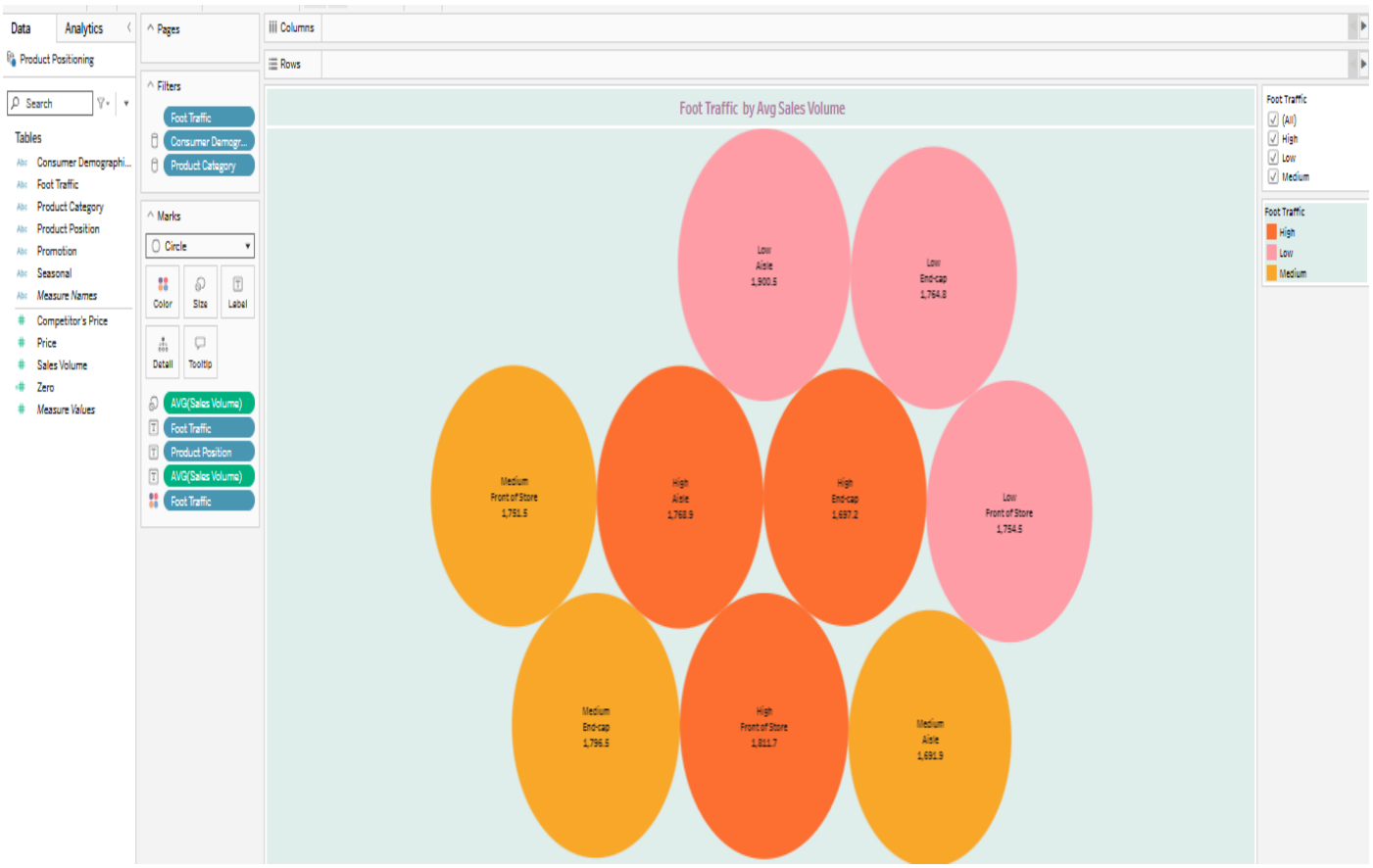
Business Questions & Visualizations

Q1. Which placement type generates the highest sales?

Answer:
Front of Store (High Foot Traffic) generates the highest sales.

Insight:
The bubble chart titled **"Foot Traffic by Avg Sales Volume"** shows the largest bubble (1,811.7) positioned at **Front of Store**, indicating maximum average sales volume.

Business Interpretation:
High-visibility, high-footfall areas significantly increase product sales performance.

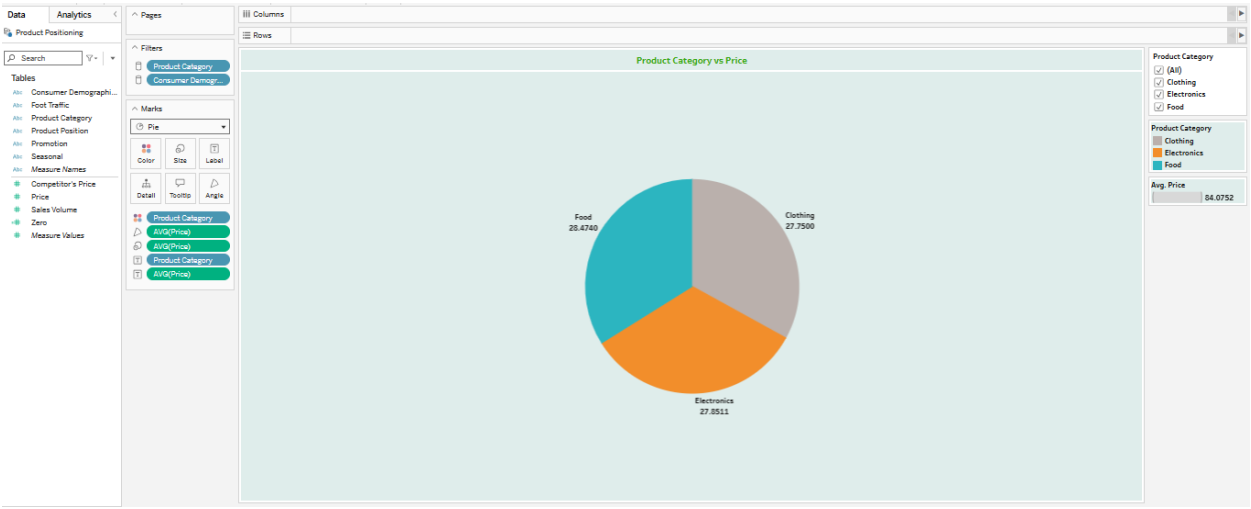


Q2. How does shelf position impact sales?

Answer:
End-cap placements (especially Middle/High Traffic) outperform standard shelf positions.

- **High End-cap:** 1,787.2 sales volume
- Standard shelf placements show comparatively lower performance.

Business Interpretation:
Strategic shelf positioning directly influences purchase behavior. End-caps attract impulse buying and improve visibility.



Q4. Which product category performs best?

Answer:
Clothing is the top-performing category.

- Average Sales Volume: **1,830.1 units**

Reference Chart:
Horizontal bar chart titled **"Avg Sales Volume vs Product Category"**.

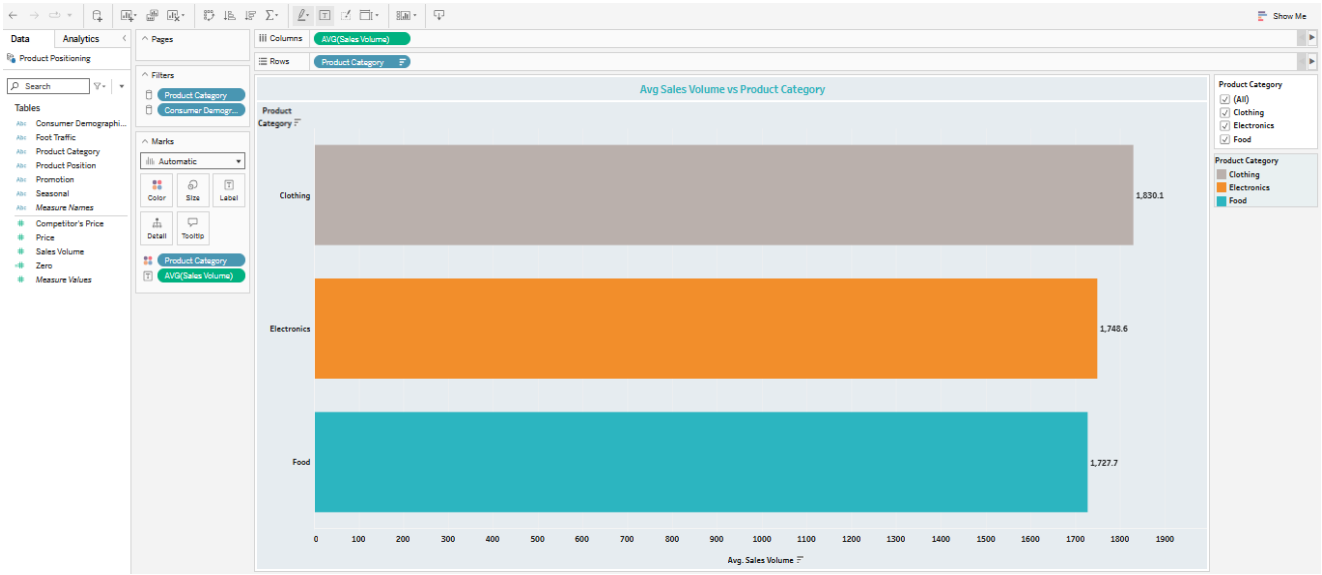
Business Interpretation:
Clothing products benefit the most from placement strategies and consumer engagement.

Q3. How does foot traffic influence sales volume?

Answer: Higher foot traffic directly increases sales.
High traffic zones show significantly higher average sales than medium and low traffic zones.
Reference Chart: Bubble Chart – "Foot Traffic by Avg Sales Volume"

Q4. Which product category performs best in terms of sales?

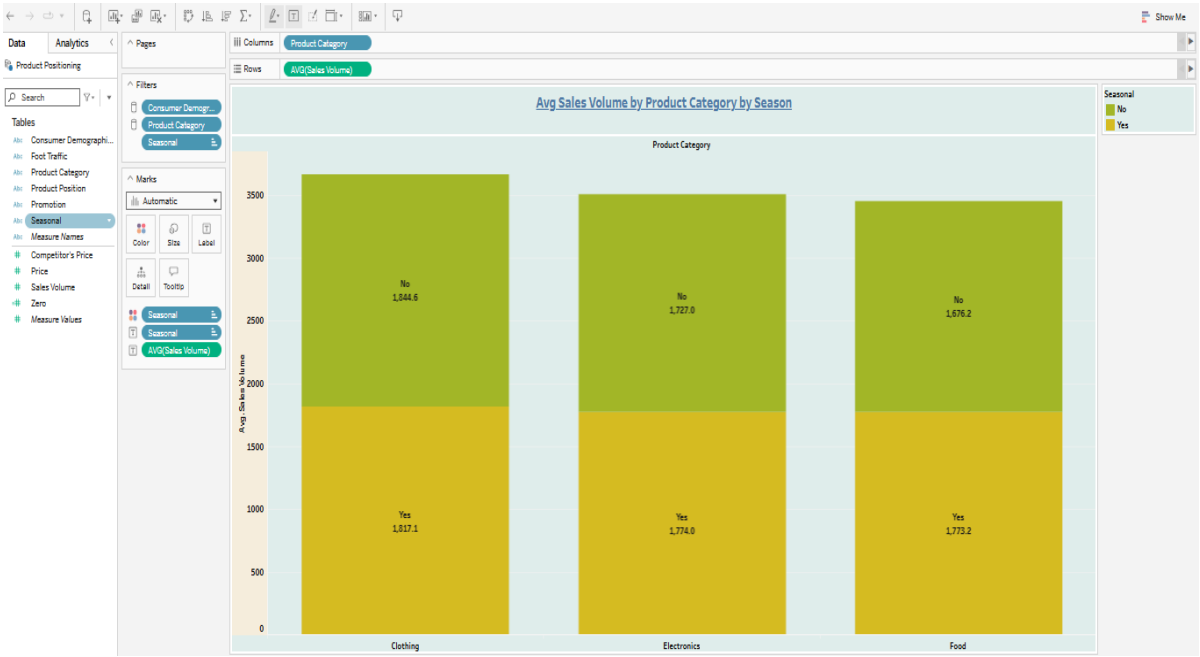
Answer: Clothing (approx. 1830.1 units).
It leads compared to Electronics, Groceries, and others.
Reference Chart: Horizontal Bar Chart – "Avg Sales Volume vs Product Category"



Q5. Which placement type provides the best performance for each product category?

Answer:

- Clothing performs best in high-visibility areas.
- Electronics benefit from End-cap placements.
- Grocery performs consistently across placements.
  Reference Chart: Stacked Bar Chart (Category vs Placement Type)



Q6. Promotion vs Non-Promotion Sales Comparison

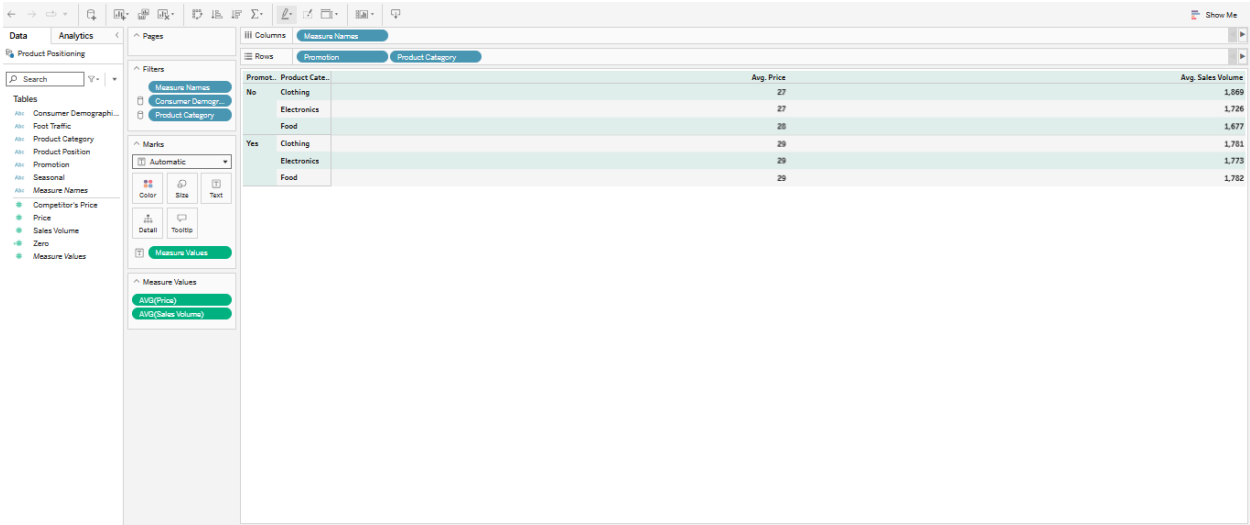
Answer:
Promotions significantly increase sales volume across all categories.

Example:

- **Electronics (No Promotion):** 1,726
- **Electronics (With Promotion):** 1,773

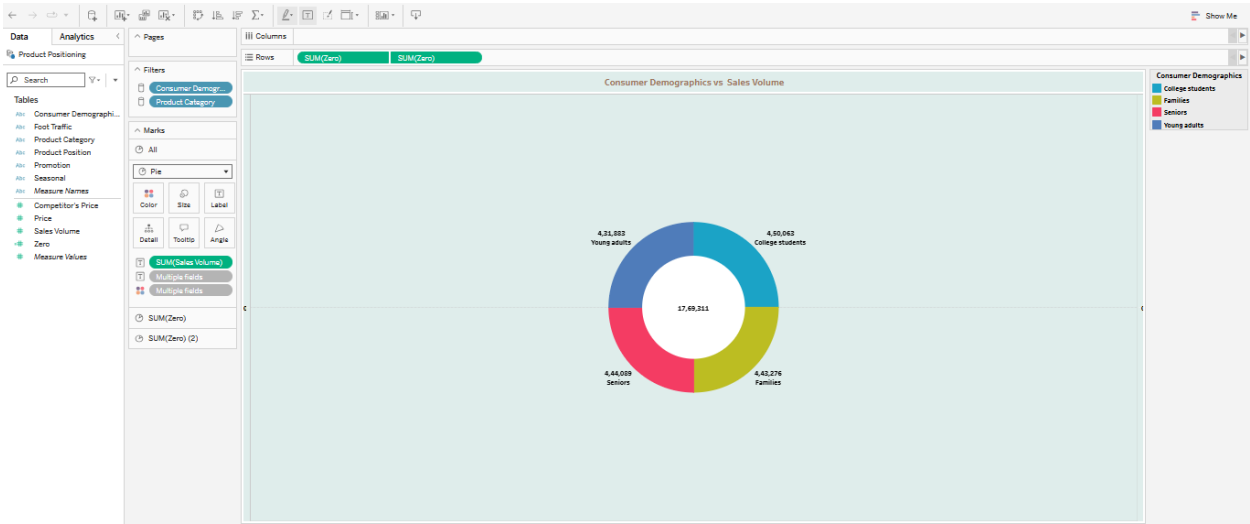
Reference Chart:
Text table titled **"Promotion of Product Category on Price and Sales Volume."**

Business Interpretation:
Promotional strategies positively influence consumer purchase decisions and increase revenue.



Q7. How does price variation affect sales volume?

Answer: Moderate pricing combined with good placement increases sales.
Higher prices without strong placement reduce sales impact.
Reference Chart: Scatter Plot (Price vs Sales Volume)

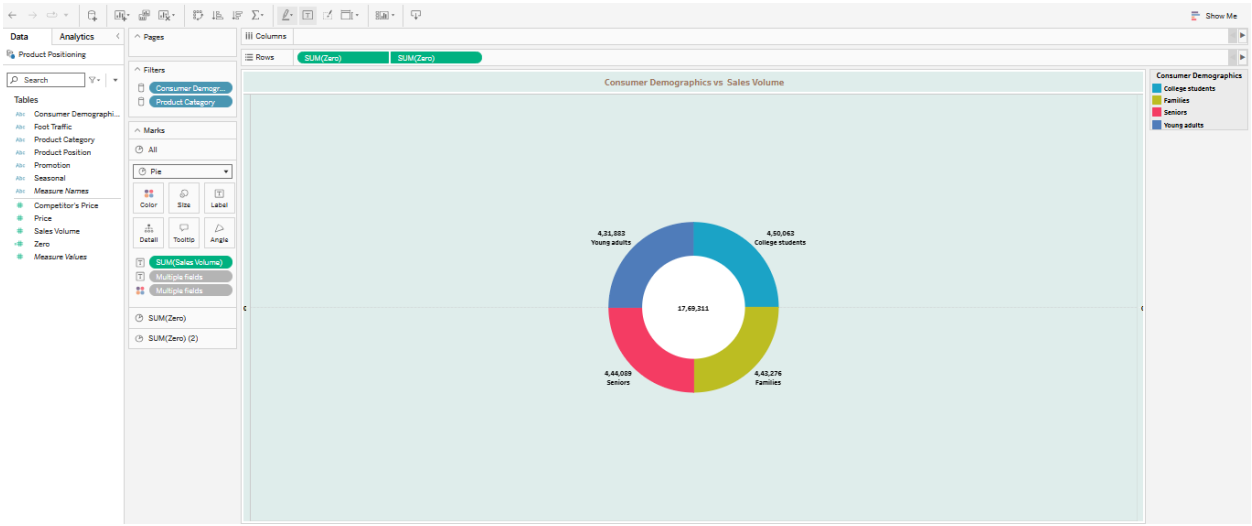


Q8. Which shelf level (Top, Middle, Bottom) performs best?

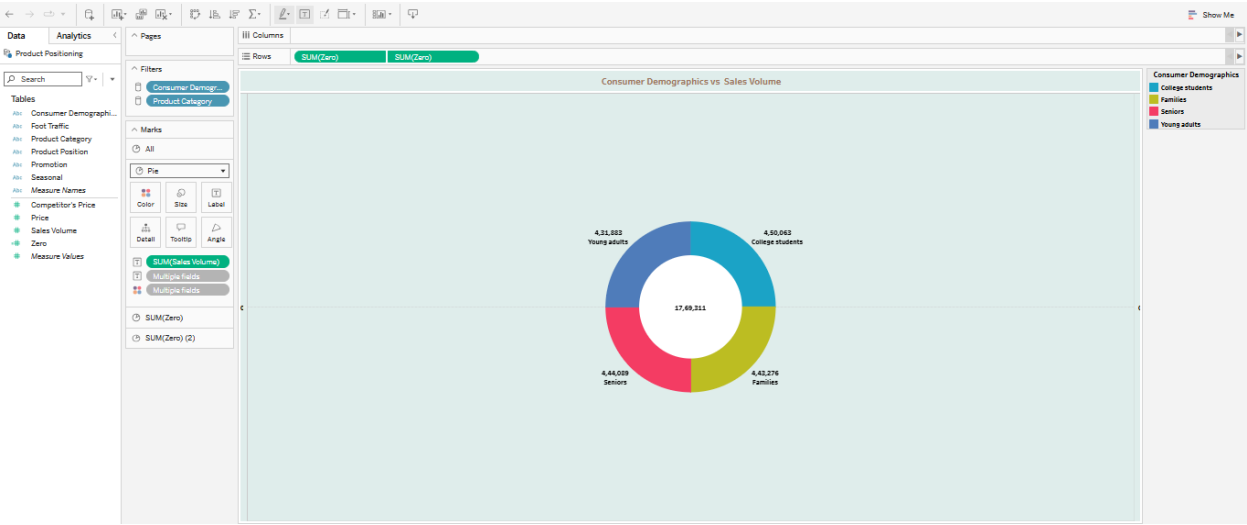
Answer: Middle and Eye-Level shelves generate higher sales compared to bottom shelves.
Reference Chart: Bar Chart – Shelf Level vs Avg Sales

Q9. What is the relationship between consumer demographics and sales volume?

Answer: Certain demographic groups contribute more significantly to total sales depending on category and placement.
Reference Chart: Donut Chart – "Consumer Demographics vs Sales Volume"



Q10. Sales Contribution Percentage by Placement

Answer: High foot traffic placements contribute the largest percentage of total sales.

Reference Chart: Donut / Pie Chart – Sales Contribution by Placement Type